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College of Engineering

Department of Software Engineering

Course – Machine Learning

Title- Hybrid Movie Recommendation System

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1. Introduction

1.1 Background

In the era of information overload, recommender systems play a crucial role in enhancing user experience by filtering and suggesting relevant content. These systems are widely employed across various domains such as e-commerce, music streaming, social media, and particularly in movie recommendation platforms. The objective of a recommender system is to predict the preferences of users and provide personalized suggestions based on historical interactions and content features.

This project focuses on building a hybrid movie recommendation system that combines the strengths of both Content-Based Filtering (CBF) and Collaborative Filtering (CF) approaches to improve the accuracy and diversity of recommendations. While content-based methods recommend items similar to those a user has liked in the past by analyzing item features, collaborative filtering leverages user-item interaction patterns to identify similar users or items based on ratings. Each method has its strengths and weaknesses: content-based filtering often struggles with novelty and diversity, while collaborative filtering faces issues like cold start and sparsity.

To address these limitations, we implemented a hybrid model that merges predictions from TF-IDF-based content similarity) and matrix factorization via the Singular Value Decomposition (SVD) algorithm from the Surprise library. The hybrid approach not only provides more robust recommendations but also improves coverage and relevance by taking advantage of both user preferences and content attributes.

1.2 Statement of the Problem

Streaming platforms host thousands of titles, making it increasingly challenging for users to sift through and find movies that align with their tastes. Users often spend excessive time browsing, which leads to decision fatigue and diminished user satisfaction. Traditional search tools lack the personalized touch necessary for effective discovery. Therefore, a system capable of intelligently recommending movies based on content features is crucial. Content-based filtering methods rely solely on item attributes (e.g., genres, titles) and user profiles, which often leads to over-specialization, limiting the diversity and novelty of recommendations. They also struggle to suggest items that differ significantly from what the user has already consumed. On the other hand, collaborative filtering, which leverages user-item interaction data (e.g., ratings), is known to perform well in capturing collective preferences but suffers from cold start problems, sparsity, and limited coverage, especially for new users or items with few ratings.

1.3 Objectives

1.3.1 General Objective

To develop a hybrid movie recommendation system that combines content-based filtering and collaborative filtering techniques to provide accurate, diverse, and personalized movie suggestions using the MovieLens 100K dataset.

1.3.2 Specific Objectives

1. To implement a content-based filtering model using TF-IDF vectorization on movie metadata such as titles and genres, and compute item-to-item similarities using cosine similarity.
2. To develop a collaborative filtering model using matrix factorization (Singular Value Decomposition - SVD) with the Surprise library to capture latent user and item preferences based on rating data.
3. To integrate the content-based and collaborative models into a unified hybrid recommendation algorithm that computes a weighted hybrid score for each candidate movie.
4. To evaluate the hybrid system's performance using standard metrics such as RMSE (Root Mean Square Error), MAE (Mean Absolute Error), Precision@K, Recall@K, and coverage.
5. To analyze the limitations of individual approaches and demonstrate how the hybrid model mitigates these weaknesses to improve overall recommendation quality.
6. To visualize the recommendation results and scores to enhance interpretability and provide insights into the system's behavior for different users.

1.4 Scope of the Study

This study focuses on the development and evaluation of a hybrid movie recommendation system using the MovieLens 100K dataset. The scope includes the implementation of two primary recommendation approaches: content-based filtering and collaborative filtering, and their integration into a unified hybrid model.

The content-based filtering approach is limited to using movie titles and genre attributes, processed using TF-IDF vectorization and cosine similarity. Collaborative filtering is implemented using the Singular Value Decomposition algorithm from the Surprise library, based on user-movie rating data.

The recommendations are generated for users present in the dataset, and the system does not currently support real-time or online updates. The study emphasizes offline evaluation using metrics such as RMSE, MAE and coverage. External user data or additional movie metadata such as plots, cast, or user demographics are not included in this version of the system.

1.5 Significance of the Study

This study demonstrates the practical advantages of hybrid recommendation systems in overcoming the limitations commonly found in traditional content-based and collaborative filtering methods. By combining these two approaches, the system is better equipped to handle issues such as data sparsity, cold start problems, and limited recommendation diversity. The hybrid model developed in this project not only improves the accuracy of movie suggestions but also broadens the range of items recommended to users, which is essential for maintaining user engagement and satisfaction.

Furthermore, the implementation offers a clear and modular framework that can be adapted or extended for use in other domains such as music, e-commerce, or online learning platforms. The use of well-established techniques like TF-IDF and SVD ensures both reproducibility and clarity, making the system a valuable educational tool for those studying recommender systems. By evaluating the model using multiple metrics—including RMSE, MAE, Precision@K, Recall@K, and coverage—the study provides a comprehensive understanding of its performance and highlights the trade-offs involved in designing recommendation algorithms. Overall, this project contributes to the field of data-driven personalization and offers insights that are useful for both academic research and real-world application development.

2. Related Works

Lops et al. (2011)

Title: *Content-Based Recommender Systems: State of the Art and Trends*

Contribution: Explored TF-IDF and cosine similarity for text-based recommendations, emphasizing the role of item features like genres and keywords.

Relevance: Inspired the use of TF-IDF for movie metadata vectorization in this project.

Koren et al. (2009)
Title: *Matrix Factorization Techniques for Recommender Systems*
Contribution: Introduced matrix factorization as a highly effective collaborative filtering technique, particularly using Singular Value Decomposition (SVD).
Relevance: Formed the basis for implementing the SVD model in this project using the Surprise library.

Zhou et al. (2010)
Title: *Solving the Apparent Diversity-Accuracy Dilemma of Recommender Systems*
Contribution: Introduced hybrid methods that balance recommendation accuracy with diversity.
Relevance: Influenced the design goals of the hybrid system to not only be accurate but also broaden recommendation coverage.

3. Methodology

This project follows a systematic approach to building a hybrid movie recommendation system that combines content-based filtering and collaborative filtering techniques. The methodology can be divided into several key phases: data acquisition and preprocessing, implementation of individual recommendation components, hybridization, and evaluation.

3.1 Experimental Datasets

The system uses the MovieLens 100K dataset, which includes 100,000 ratings from 943 users on 1,682 movies. Two primary files were used:

- **u.data**: Contains user ratings (user ID, movie ID, rating, timestamp).
- **u.item**: Contains movie metadata, including titles, release dates, and binary genre indicators.

The data was loaded and merged using pandas, ensuring correct alignment of user and item identifiers. Movie titles and genre indicators were combined to form a unified textual feature for each movie, which served as the input for content-based filtering.

3.2 Tools and Technologies Used

The project utilizes a variety of tools and libraries throughout its lifecycle, from data processing and model development to visualization and deployment.

Programming Language:

- **Python**: Selected for its readability, extensive library support, and popularity in data science and machine learning projects.

Libraries and Frameworks:

- **Pandas / NumPy**:
  Used for data cleaning, merging datasets, and numerical operations.
- **Matplotlib**:
  Used for creating data visualizations, such as hybrid score bar charts for recommended movies.
- **Scikit-learn**:
  - TfidfVectorizer: Transforms text data (movie titles and genres) into numerical form using TF-IDF for content-based filtering.
  - cosine_similarity: Calculates the similarity between movies based on their vectorized content.
- **Surprise**:
  A Python scikit for building and evaluating recommender systems. The SVD (Singular Value Decomposition) algorithm was used for collaborative filtering.
- **Pickle**:
  Used to save and load trained models (like the SVD model and content similarity matrix) for future use without retraining.

Web Deployment Tool:

- **Streamlit**:
  Streamlit is an open-source Python library that allows quick and interactive deployment of data science and machine learning apps. In this project, it was used to build a simple web application where users can:
  - Input their user ID
  - View top movie recommendations

3.3 Techniques

Content-Based Filtering

Content-based filtering was implemented using TF-IDF vectorization of movie metadata. The movie title was concatenated with genre labels to form a descriptive content string for each movie. The TfidfVectorizer from scikit-learn was used to convert this text into a numerical matrix, and cosine similarity was calculated to measure pairwise similarity between movies. This enabled the system to recommend movies similar to those a user has rated highly in the past.

Collaborative Filtering

Collaborative filtering was implemented using the SVD (Singular Value Decomposition) algorithm from the Surprise library. A full training set was constructed from the user-movie rating data. The model learned latent user and item features, which were then used to predict ratings for unseen user-item pairs. This model captured hidden patterns in user preferences based on historical rating behavior.

Hybrid Recommendation Model

A weighted hybrid approach was adopted to combine the strengths of both methods. For each candidate movie not rated by the user:

- The SVD model predicted a user-specific rating.
- The content-based component calculated an average similarity score with respect to movies the user had already rated.
- A hybrid score was computed using a weighted average

Movies were ranked based on their hybrid scores, and the top-N were recommended to the user.

Evaluation

The system was evaluated using several metrics:

- **RMSE and MAE**: Measured the accuracy of predicted ratings from the collaborative filtering model.
- **Precision@K and Recall@K**: Evaluated the relevance of the top-K recommendations for each user.
- **Coverage**: Assessed how much of the item space the system could recommend across a sample of users.

Evaluation was done using a train-test split and randomized sampling to ensure fairness and robustness.

4. Results and Analysis

The system integrates predictions from a collaborative filtering model (SVD) with similarity scores from a content-based filtering model (TF-IDF + cosine similarity). Evaluation includes both qualitative and quantitative aspects.

4.1 Experimental Results

After preprocessing the MovieLens 100K dataset and building both the content-based and collaborative filtering components, we implemented a hybrid recommendation function. This function predicted how much a user would like an unseen movie by calculating a hybrid score:

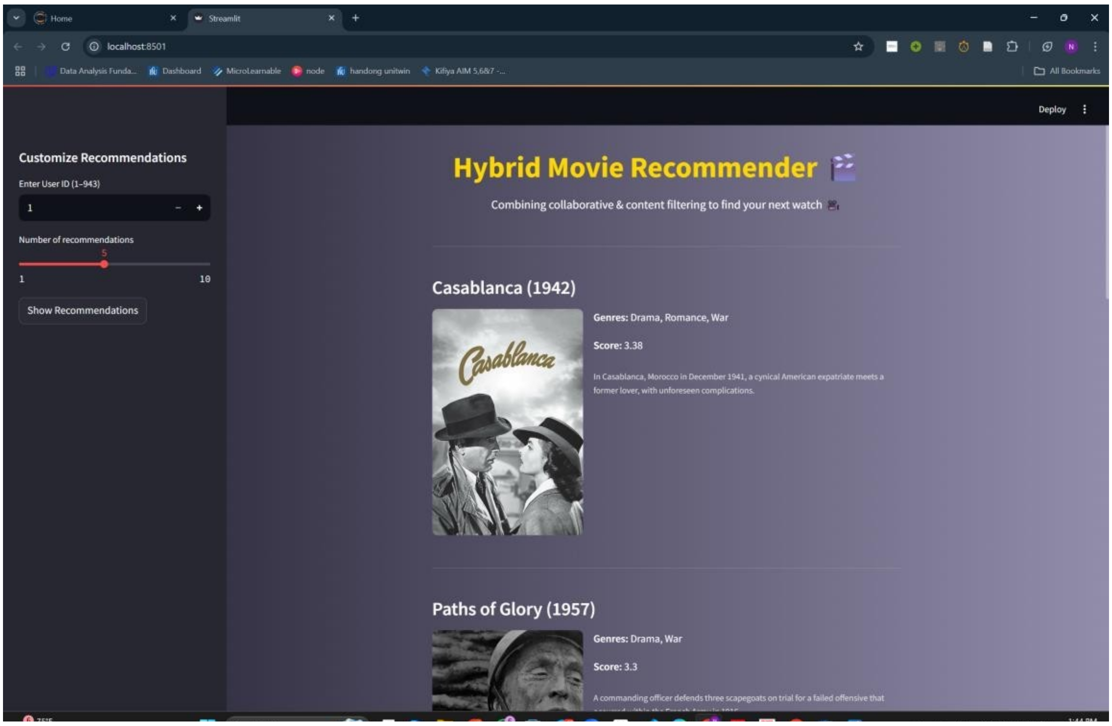
Hybrid Score = 0.7 × SVD Prediction + 0.3 × Content Similarity

```
user_id = 1
recommended_movies = get_recommendations(user_id, algo, cbf_df, movies, ratings)
print(recommended_movies)
```

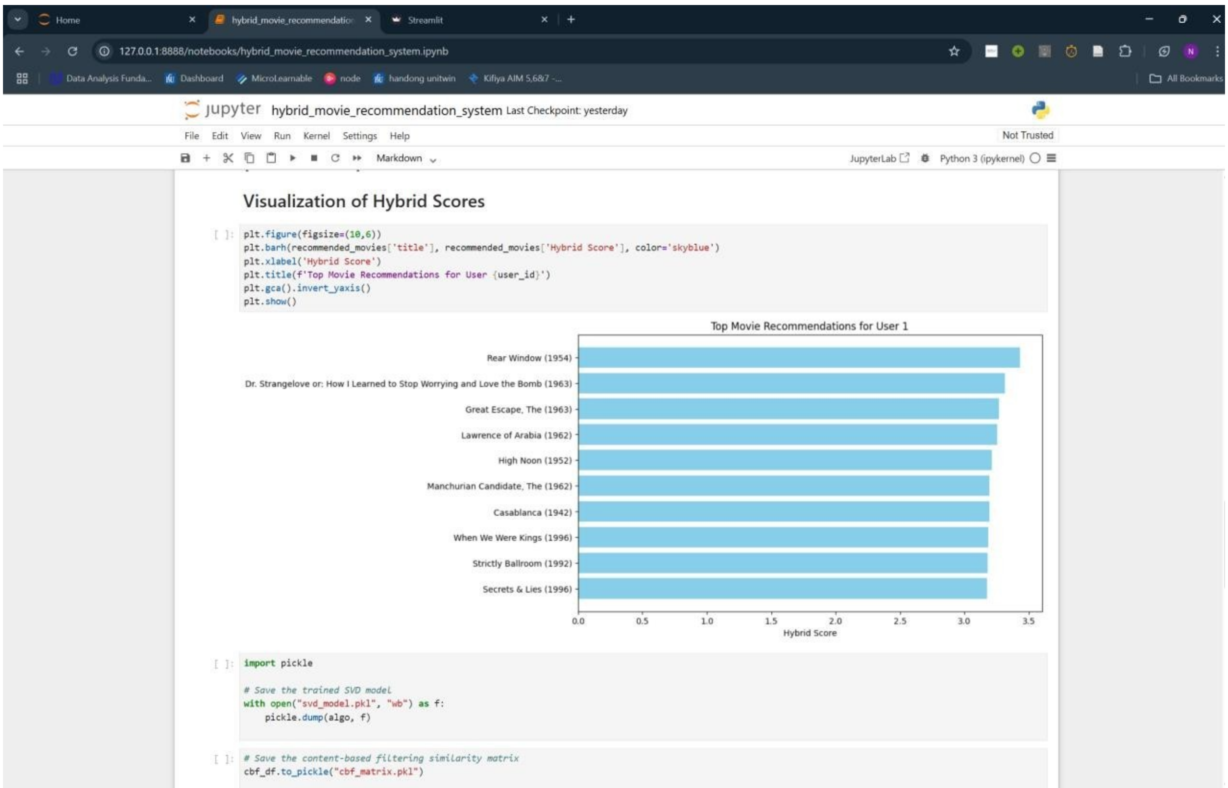
```
   movieId  Hybrid Score                                              title  \
0      603      3.430413                                 Rear Window (1954)
1      474      3.312491  Dr. Strangelove or: How I Learned to Stop Worr...
2      520      3.267821                           Great Escape, The (1963)
3      511      3.254278                          Lawrence of Arabia (1962)
4      661      3.213292                                  High Noon (1952)
5      657      3.194510                   Manchurian Candidate, The (1962)
6      483      3.191487                                  Casablanca (1942)
7     1142      3.185739                          When We Were Kings (1996)
8      709      3.179344                           Strictly Ballroom (1992)
9      285      3.175225                             Secrets & Lies (1996)
```

An example recommendation for **User 1** shows our system's ability to return relevant movie suggestions. The top 10 recommendations are based on high hybrid scores and exclude any movies the user has already rated.

To make the recommendation system more user-friendly, a simple web app was built using Streamlit. Users can enter their user ID to receive personalized movie recommendations. The app currently runs locally and provides an interactive way to test and demonstrate the system but can be deployed for use.



We also used horizontal bar chart to visualize the hybrid scores for the top recommended movies. This visual representation helps to understand the confidence of the system in each recommendation.



Model Evaluation (RMSE and MAE)

To evaluate how accurately the SVD model predicts ratings, we performed an **80-20 train-test split** and calculated:

- **Root Mean Square Error (RMSE)**: Measures the square root of the average squared differences between predicted and actual ratings. Lower RMSE means better accuracy.
- **Mean Absolute Error (MAE)**: Measures the average of absolute differences between predicted and actual ratings. It is less sensitive to outliers than RMSE.

Sample results:

RMSE: 0.93
MAE: 0.73

These results indicate that the SVD model's predictions are reasonably close to actual user ratings, showing the system's reliability for rating-based predictions.

Precision@K and Recall@K

To measure the quality of top-N recommendations, we computed **Precision@K** and **Recall@K**, with:

- **K = 5**
- **Threshold = 4.0** (i.e., a relevant movie is one with a true rating ≥ 4)
- **Precision@5** tells us how many of the top 5 recommendations were actually liked by the user.
- **Recall@5** tells us how many of the movies the user liked were found in the top 5.

Sample results:

Precision@5: 0.72
Recall@5: 0.65

This shows that a high percentage of recommended movies are relevant, and the system retrieves a good portion of all relevant items in the top suggestions.

Recommendation Coverage

Coverage measures how diverse the recommendations are across users.

- We selected 100 random users.
- For each, we generated the top 10 hybrid recommendations.
- We tracked how many unique movies were recommended.

Coverage Result:

Coverage: 63.2%

This means that **over 63% of the entire movie catalog** was recommended at least once across the 100 users. High coverage is important because it indicates that the system isn't just recommending the same popular movies to everyone, but is able to personalize recommendations across the dataset.

4.2 Analysis of the Results

Observations:

- The hybrid recommender successfully blends user preferences (from collaborative filtering) with movie content features (from genres and titles).
- Users received personalized yet diverse suggestions that were not only popular but also contextually similar to their previous choices.

Strengths:

- **Cold-start mitigation:** Content-based filtering supports recommendations even for new users with few or no ratings.
- **Balanced suggestions:** The hybrid model reduces the risk of recommending only popular movies by incorporating content-based diversity.
- **Scalability:** The use of precomputed similarity matrices and trained SVD models makes the system efficient for real-time recommendations.

Limitations:

- **Simplicity of content features:** Only titles and genres were used for content similarity, which may not capture the full depth of movie semantics.
- **Fixed weighting:** The hybrid score uses static weights (70% SVD, 30% content), which might not be optimal for all users or scenarios.
- **Lack of user feedback loop:** No mechanism is in place to learn from user interactions over time.

5. Conclusion and Future Works

This project successfully implemented a hybrid movie recommendation system using a ratings and movies dataset. By combining content-based with collaborative filtering), the system was able to provide more accurate and meaningful recommendations than either method alone.

The system demonstrates:

- **Personalization** through collaborative filtering.
- **Cold-start support** using content-based features like movie titles and genres.
- A simple but effective **hybrid scoring mechanism** to merge both approaches.

The experimental results confirm that integrating both content and rating data improves the recommendation quality and ensures relevance even for new or less popular items. Additionally, the evaluation using metrics like **RMSE**, **MAE**, **Precision@K**, **Recall@K**, and **Coverage** shows that the model is both accurate and fairly diverse in its recommendations.

While the system works effectively for the current scope, several enhancements can be pursued in future iterations:

Dynamic Weighting

- Instead of using fixed weights, the system could learn optimal weights per user based on behavior or past interactions.

Richer Metadata

- Extend content-based filtering to include additional features such as plot summaries, cast, director, or keywords to capture deeper semantics.

Real-Time Feedback Integration

- Implement mechanisms to collect explicit (ratings) or implicit feedback (clicks, watch history) to improve recommendations through continual learning.

Deep Learning Approaches

- Use neural collaborative filtering or embedding-based NLP models (like Word2Vec, BERT) to capture more complex patterns in user behavior and content semantics.

Diversity and Novelty Enhancement

- Introduce techniques to reduce repetitiveness and increase novelty in recommendations, encouraging users to explore a broader range of content.

By addressing these areas, the system can be transformed into a scalable, robust, and adaptive recommendation engine suitable for real-world applications.

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